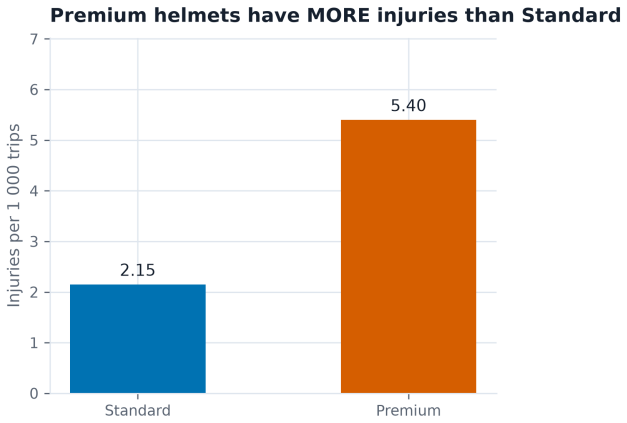
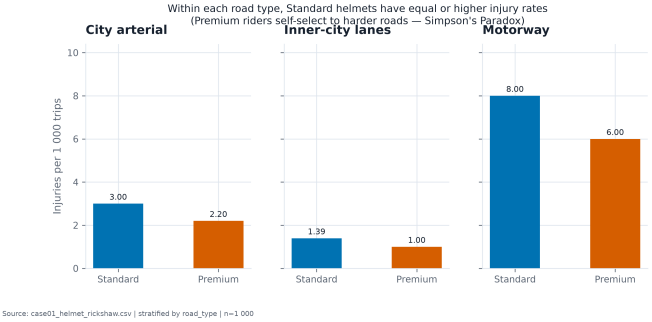
Part A — Nine Diagnoses

Week 3 Data Visualisation Assignment | Roll No: BSAI-010

1. Case 01 Helmets



Source: case01_helmet_rickshaw.csv | n=1 000 riders | 100 trips each



Source: case01_helmet_rickshaw.csv | stratified by road_type | n=1 000

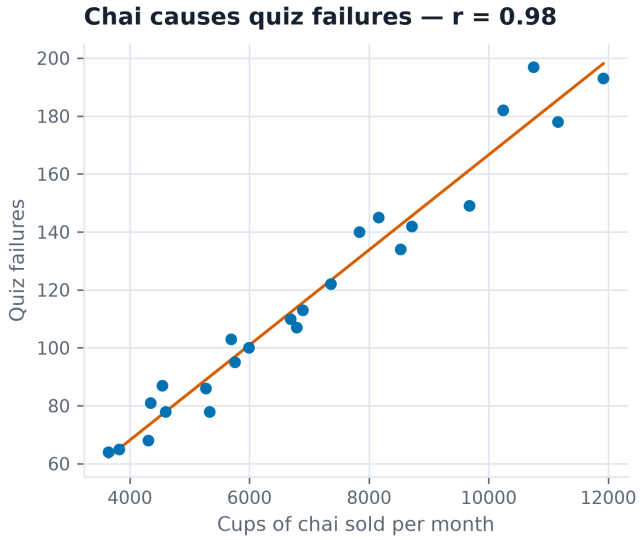
Invited conclusion: Premium helmets cause more injuries than Standard helmets.

Decoy objection ruled out: Could be rider skill? Ruled out: experience months are similar across grades (mean ~40).

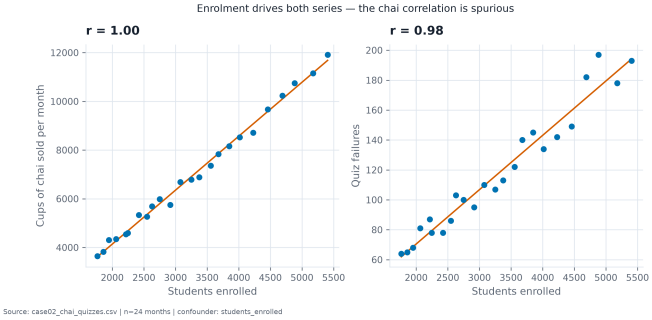
Mechanism: Simpson’s Paradox / confounding by road type. Premium riders concentrate on Motorways (higher absolute risk); within each road type, Standard helmets show equal or higher rates.

Redesign: Bar chart faceted by road_type — removes the confound and reverses the conclusion.

2. Case 02 Chai–Quizzes



Source: case02_chai_quizzes.csv | n=24 months



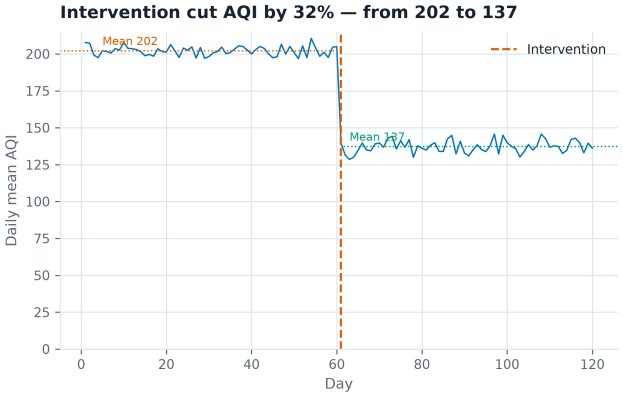
Invited conclusion: Drinking chai causes quiz failures (r = 0.98).

Decoy objection ruled out: Could be exam-month effect? Ruled out: is_exam_month has only 3 flagged months.

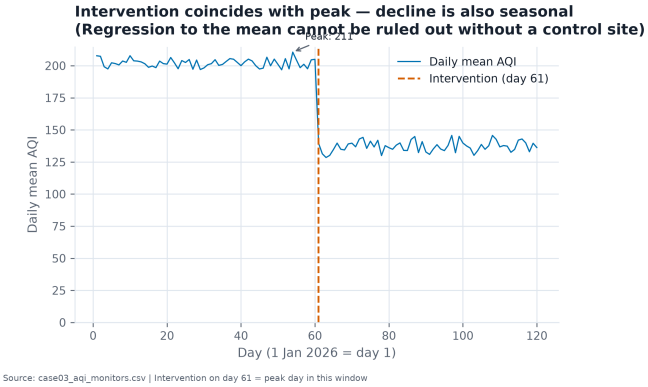
Mechanism: Spurious correlation driven by the omitted confounder students_enrolled (r=0.99 with chai sold, r=0.97 with failures). Enrolment grows monotonically and drives both series.

Redesign: Two side-by-side scatter plots of each series vs enrolment — the chai↔failures relationship disappears after conditioning.

3. Case 03 Lahore AQI



Source: case03_aqi_monitors.csv | daily mean across stations



Source: case03_aqi_monitors.csv | Intervention on day 61 = peak day in this window

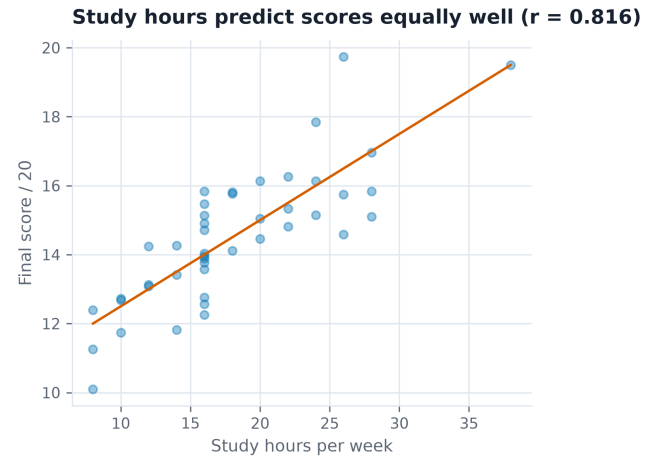
Invited conclusion: The day-61 intervention cut AQI by 32% (202→137).

Decoy objection ruled out: Could be seasonal? The window is only 120 days, but the AQI peaks on day ~55 and declines regardless — consistent with a winter-to-spring seasonal effect.

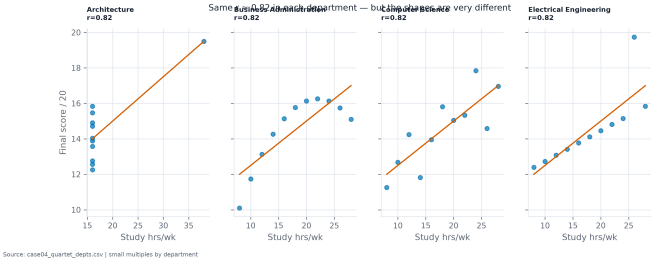
Mechanism: Regression to the mean: the intervention was announced at an AQI peak. Any measurement taken at a peak will tend to fall; no control site is provided to distinguish the policy from the season.

Redesign: Full line chart annotating both the intervention date and the seasonal peak, and noting the absence of a control.

4. Case 04 Quartet Depts



Source: case04_quartet_depts.csv | all departments pooled



Source: case04_quartet_depts.csv | small multiples by department

Invited conclusion: Study hours predict scores equally well across all departments (r ≈ 0.82).

Decoy objection ruled out: Could department size explain the difference? No — n is similar across groups.

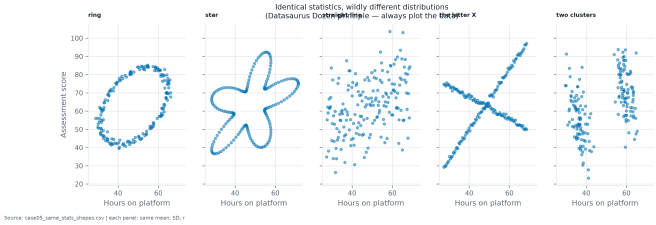
Mechanism: Anscombe’s Quartet: identical summary statistics (mean, SD, r) can mask radically different distributions. The scatter plots reveal linear, curved, outlier-dominated, and step-function patterns.

Redesign: Small-multiple scatter plots — one per department — with OLS lines.

5. Case 05 Same Stats

All cohorts share identical statistics — same platform effect?

Shape	Mean hrs	Mean score	SD hrs	SD score	r
ring	48.00	63.00	12.00	15.00	0.420
star	48.00	63.00	12.00	15.00	0.420
straight line	48.00	63.00	12.00	15.00	0.420
the letter X	48.00	63.00	12.00	15.00	0.420
two clusters	48.00	63.00	12.00	15.00	0.420



Source: case05_same_stats_shapes.csv | summary statistics only

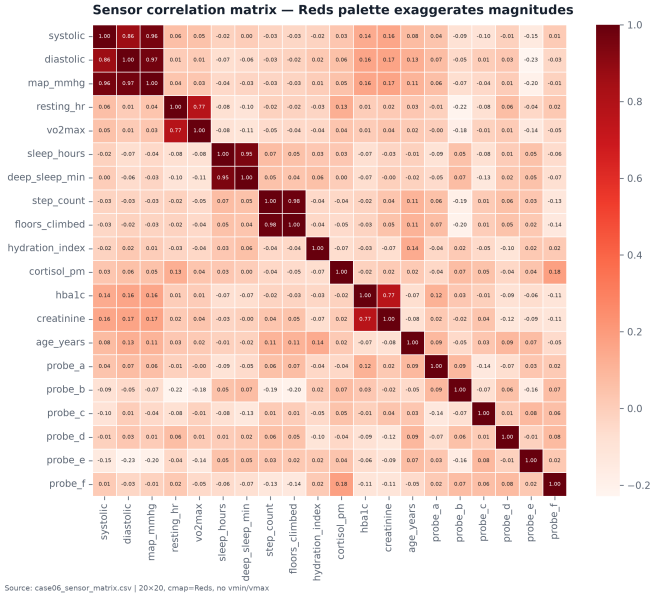
Invited conclusion: All five cohorts respond identically to the platform (same stats).

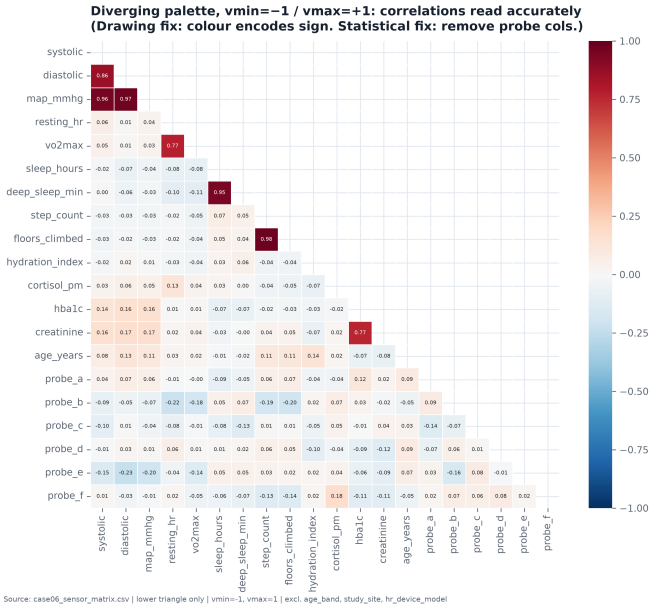
Decoy objection ruled out: Could the equal stats be a coincidence? No — they are exact by construction.

Mechanism: Datasaurus Dozen / match_stats(): summary statistics do not constrain shape. r = 0.42 is consistent with a straight line, a star, a circle, or noise.

Redesign: Small-multiple scatter plots showing wildly different shapes with identical statistics.

6. Case 06 Sensor Matrix





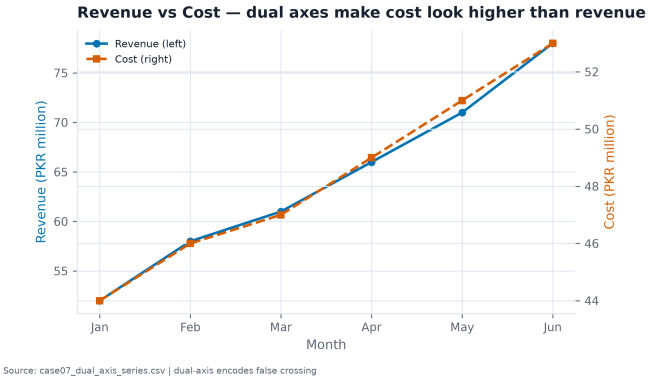
Invited conclusion: All sensors are strongly correlated — the matrix is uniformly warm (Reds).

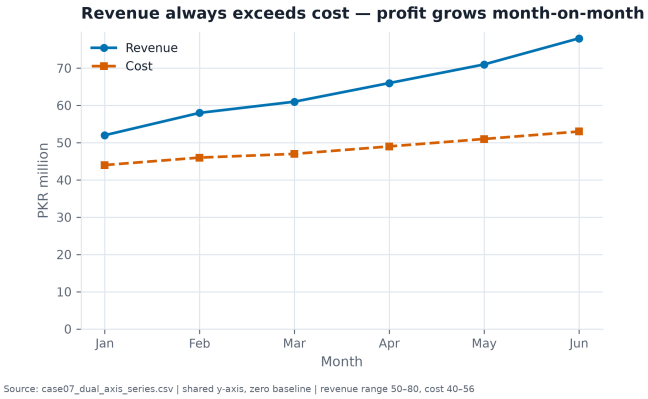
Decoy objection ruled out: Could the high correlations be real? Partially: systolic–MAP 0.96 is physiologically expected. But probe columns (probe_a–f) are artefactual.

Mechanism: Two problems: (1) Statistical — probe columns are instrument noise, not physiology; including them inflates apparent structure. (2) Drawing — cmap='Reds' without vmin/vmax maps correlation 0 to white and 1 to red, so a weak 0.2 looks dark. The drawing fix (diverging palette, vmin=-1) solves problem (2) only; the statistical fix requires dropping probe columns.

Redesign: Lower-triangle heatmap, cmap=RdBu_r, vmin=-1, vmax=1 — correlations read as magnitudes and signs correctly.

7. Case 07 ShaheenGo





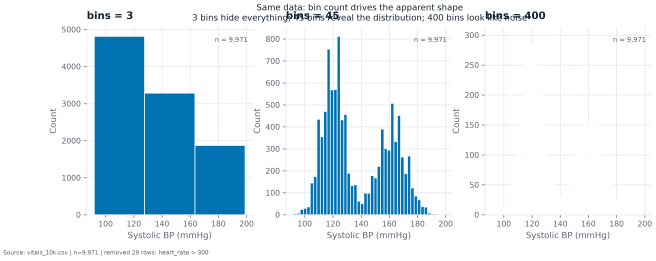
Invited conclusion: Cost exceeds revenue in the first four months.

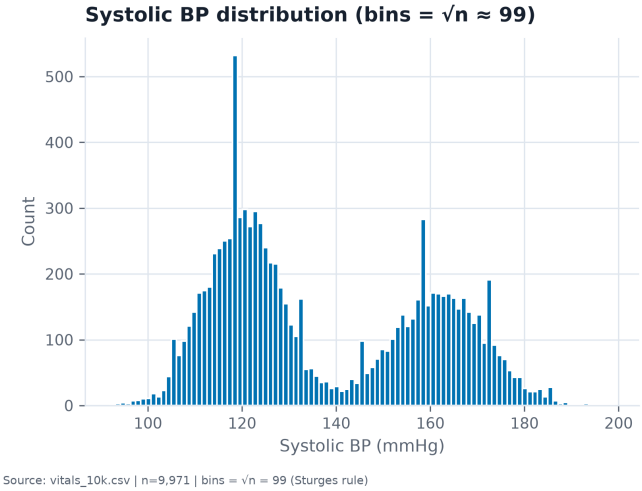
Decoy objection ruled out: Could the raw numbers confirm this? No — revenue (50–80) always exceeds cost (40–56).

Mechanism: Dual-axis encoding: the designer chose scale ranges that made the cost line plot above the revenue line even though cost < revenue every month. Any crossing point can be manufactured by adjusting one axis independently.

Redesign: Single shared axis, zero baseline — cost and revenue never cross.

8. Case 08 Histograms





Source: vitals_10k.csv | n=9,971 | bins = √n = 99 (Sturges rule)

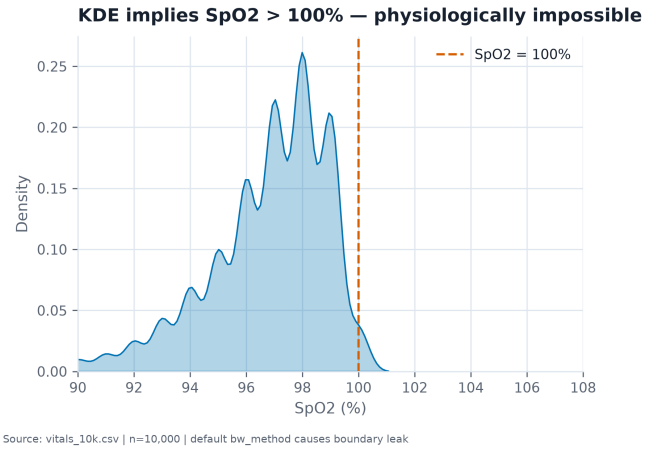
Invited conclusion: The distribution of systolic BP is unimodal (3 bins) or bimodal (400 bins).

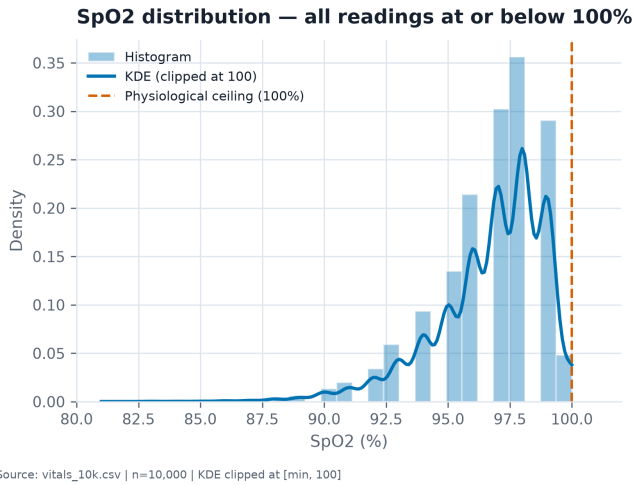
Decoy objection ruled out: Could the bimodality be real? Ruled out: 45-bin histogram shows a single right-skewed distribution with no gap.

Mechanism: Bin-count sensitivity: 3 bins destroy all shape information; 400 bins amplify sampling noise into apparent secondary peaks. The √n rule (≈ 100 bins) balances detail against noise.

Redesign: Single histogram with bins = √n, bin width declared in the axis label.

9. Case 09 SpO2 KDE





Invited conclusion: Some patients have SpO2 > 100% (the KDE extends to ~103%).

Decoy objection ruled out: Could the device capture >100%? No — the data maximum is exactly 100.

Mechanism: Gaussian KDE has infinite support: it always leaks beyond the data boundary. Here the boundary is a hard physiological ceiling (100%), so any mass above it represents impossible values. The fix is clip=(min, 100) in kdeplot.

Redesign: Overlaid histogram + KDE clipped at 100, with a vertical reference line.